

INTERACTIONS EXPLORED USING AN EXTENSION OF PROFILE ANALYSIS

Ross Sparks and Roy Ballantyne	Department of Mathematics, University of Wollongong, Wollongong N.S.W., Australia Department of Environment and Geographical Sciences University of Cape Town, South Africa
---	--

Keywords:	Repeated measures, MANOVA, design, hypothesis testing, model selection
------------------	--

Summary:	The traditional profile analysis is extended to repeated measures on several response variables. A modeling approach is used to describe interactions in the analysis. Model selection and hypothesis testing procedures are emphasised in order to decide whether variables and/or factors interact. The application used to illustrate the methodology examines six assessors' rating of identical performances in three question-answers.
-----------------	--

1. INTRODUCTION

Traditionally profile analysis has been carried out using repeated measures of one response variable (see for example: Mardia et al, 1979; Morrison, 1990; Srivastava and Carter, 1983 and Srivastava, 1987). Although the technique is useful in these situations, different designs are required when examining repeated measures of several response variables or where the nature of possible complex inter-relationships need to be explored. The methodology can, however, be extended to designs with several factors (and their interactions). A modelling approach will be used to illustrate interactions, while selection procedures are used to decide whether variables and/or factors interact [Linhart and Zucchini (1986)].

Although model selection procedure is emphasised, various statistical criteria may be used in testing hypotheses, viz., Roy's largest root, Hotelling's T^2 statistic, Wilk's likelihood ratio and Pillai's criterion (Morrison, 1990 and Srivastava, 1987). The exact distributions of these are unknown in the application (section 6); however, asymptotic distributions are available for large data sets. For small sample sizes, a Monte Carlo approach using "bootstrap" methodology is proposed for testing hypotheses.

In the design experiment considered in this paper there are:

- four hypotheses, known in the literature as tests for "parallel profiles";
- a hypothesis that tests whether the two "response" variables interact;
- a test whether the factor levels are equal; and
- two hypotheses that test for equal "response" means.

These are presented and will be discussed in section 2.

2. THE DESIGN AND THE OPERATING MODEL

Let y_{ijk} be an r -dimensional vector of rating scores assigned to the same performance by r assessors. The performance relates to the k 'th task (or question/essay/act) by the i 'th person from the j 'th sector of the population; $j = 1, \dots, s$; $i = 1, \dots, n_j$ and $k = 1, \dots, q$. The mean vector of assessor's scores for a person from the j 'th sector in the performance of the k 'th task is μ_{jk} (a $r \times 1$ vector). The random vector y_{ijk} are assumed multivariate normal with covariance matrix Σ such that

$$\begin{aligned} E(y_{ijk} - \mu_{jk})(y_{ij'k'} - \mu_{j'k'}) &= 0 \text{ if either } j \neq j' \text{ or } k \neq k'; \\ &= q'_k R \text{ if } j = j', i = i' \text{ and } k = k'; \\ &= q_{kk'} R \text{ if } j = j', i = i' \text{ and } k \neq k'. \end{aligned}$$

where: R is a $r \times r$ matrix representing the covariances between assessors.

Denote:

$$y'_{ij} = (y'_{ij1} \cdots y'_{ijq});$$

and

$$\mu_j = (\mu'_{j1} \cdots \mu'_{jq}).$$

The operating model for the random vector y_{ij} is considered to be

$$(1) \quad y_{ij} = \mu_j + e_{ij}$$

where: e_{ij} is a vector of random errors assumed i.i.d. $N(0, \Sigma), j = 1, \dots, n; i = 1, \dots, s;$

$$\Sigma = Q \otimes R, R = \begin{pmatrix} r_1^2 & r_{12} & \cdots & r_{1r} \\ r_{12} & r_2^2 & \cdots & r_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1r} & r_{2r} & \cdots & r_r^2 \end{pmatrix}; Q = \begin{pmatrix} q_1^2 & q_{12} & \cdots & q_{1q} \\ q_{12} & q_2^2 & \cdots & q_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ q_{1q} & q_{2q} & \cdots & q_q^2 \end{pmatrix}$$

and 0 is a $qr \times 1$ zero vector.

Equation (1) will be referred to as the operating model. There may be situations where a more general structure for Σ should be applied. For this application the above structure was *a priori* reasonable and was demonstrated to be empirically sound (the likelihood ratio test for $H_0 : \Sigma = Q \otimes R$ vs $H_a : \Sigma = \text{general structure}$ was not significant, i.e. $p > 0.2$). This assumption can be interpreted as reflecting no interaction between the scales of the questions' scores and assessors' assessments. This presumes that all assessors have their own implicit scale (spread of scores) which remains constant across all questions. The question scores are related, however, as they test overlapping content areas and/or skills.

The multivariate regression form of the operating model is

$$Y = X\theta + E$$

$$\text{where: } y = \begin{pmatrix} y'_{11} \\ \vdots \\ y'_{n_1,1} \\ \vdots \\ y'_{1s} \\ \vdots \\ y'_{n_s,s} \end{pmatrix}; X = \begin{pmatrix} 1_{n_1} & \cdots & 0_{n_1} \\ \vdots & \ddots & \vdots \\ 0_{n_s} & \cdots & 1_{n_s} \end{pmatrix}; \theta = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_s \end{pmatrix} \text{ and } E$$

is the random error component of the model assumed normal with mean equal to a zero matrix and covariance matrix $Q \otimes R$.

The maximum likelihood estimate of θ is

$$\hat{\theta} = (X'X)^{-1}X'Y.$$

A closed form solution to the maximum likelihood procedure for estimating the covariance matrices is not available, however, an iterative solution is proposed. Note, that the representation of Σ is not unique, i.e. $\Sigma = (cQ) \otimes (\frac{1}{c}R)$. Hence, in the estimation process to follow it is assumed that either R or Q will be restricted so that their diagonal elements average to one. Provided the likelihood function is unimodal, this restriction guarantees that the iterative solution to equations below converge to the same solution for various starting points. Let

$$G = (Y - X\hat{\theta})'(Y - X\hat{\theta})$$

$$(2) \quad = \begin{pmatrix} G_{11} & \cdots & G_{1q} \\ \vdots & \ddots & \vdots \\ G_{q1} & \cdots & G_{qq} \end{pmatrix}$$

where: G_{ij} are $r \times r$ matrices formed by the corresponding partitioning of matrix G . A maximum likelihood estimate can be solved iteratively from the following two equations (see Appendix A):

$$(3) \quad Q_{mle} = \frac{1}{nr} A$$

where: the i 'th row and j 'th column element of matrix A is given by $A_{ij} = \text{tr}[R^{-1}G_{ij}]$; and

$$(4) \quad R_{mle} = \frac{1}{nq} B$$

where: the i 'th row and j 'th column element of matrix B is given by $B = \sum_{i=1}^q \sum_{j=1}^q [Q^{-1}]_{ij} G_{ji}$, provided they converge to a global maximum. If the data are normal then a global maximum can be anticipated.

These maximum likelihood estimates can be used to establish test statistics for hypotheses of the form

$$(5) \quad H_0 : C\theta(V_1 \otimes V_2) = 0 \text{ vs } H_a : C\theta(V_1 \otimes V_2) \neq 0$$

where: for the application known matrices C, V_1 and V_2 are of full row rank, full column rank and full column rank respectively. The classical test statistics for the above hypothesis are functions of the eigenvalues of matrix $E^{-1}H$ where

$$H = [C\hat{\theta}(V_1 \otimes V_2)]'[C(X'X)^{-1}C']^{-1}[C\hat{\theta}(V_1 \otimes V_2)]$$

and

$$E = n^2 V_1' Q_{mle} V_1 \otimes V_2' R_{mle} V_2.$$

The nested hypothesis for the design considered in this paper is outlined in Table 1. This also presents the corresponding model for each sequence of null hypotheses being true. The models are compared using the model selection approach (Section 3). In table 1, θ is taken as a $s \times qr$ matrix of location parameters in the model. In its most general form (operating model), θ measures the effect of all interactions and individual mean influences on the response variable (assessment scores) and/or sectors. Each hypothesis in Table 1 is presented in terms of θ and matrices C_a defined by

$$C'_a = (I_{a-1} \quad -1_{a-1}).$$

However, note that the form of θ may be simplified by assuming a hierarchy of hypotheses to be true. For example, if H_0^{3w} is true then a possible form θ could take is

$$(6) \quad \theta = \mu 1'_{qr} + 1'_q \otimes \beta + \alpha \otimes 1'_r + 1_s \gamma'$$

where: μ is a $s \times 1$ vector of sector mean scores; $\beta; \alpha$ and γ are $s \times r; s \times q$ and $qr \times 1$ matrices relating to location coefficients which measure the influence of "sector-assessor"; "sector-question" and "question-assessor" interaction, respectively. However, the latter three parameters contain information on individual influences of questions and assessors. It is easy to see that θ as expressed in (6) satisfies $C'_q \theta (C_q \otimes C_r) = 0$. Hence model (ii) is a possible form for the operating model if H_0^{3w} is true (Table 1). Similarly, since model (vi) was established assuming $H_0^{3w}; H_0^{sq}$ and H_0^{sr} are true, $\theta = \mu 1'_{qr} + 1_s \gamma'$ satisfies $C'_s \theta (C_q \otimes C_r) = 0; C'_q \theta (C_q \otimes 1_r) = 0$ and

Table 1: The nested hypotheses and the corresponding approximating model

Operating model	(i) $Y = X\theta + E$		
Hypothesis assumed true	$H_0^{3w} : C_q^t \theta (C_q \otimes C_r) = 0_{(s-1) \times (q-1)(r-1)}$		
Model	(ii) $Y = X[\mu 1_{qr}^t + 1_q^t \otimes \beta + \alpha \otimes 1_r^t + 1_s \gamma^t] + E$		
Hypothesis	$H_0^{sq} : C_q^t \theta (C_q \otimes 1_r) = 0_{(s-1) \times (q-1)}$	$H_0^{sr} : C_s^t \theta (1_q \otimes C_r) = 0_{(s-1) \times (r-1)}$	$H_0^{qr} : 1_s^t \theta (C_q \otimes C_r) = 0_{1 \times (q-1)(r-1)}$
Model	(iii) $Y = X[\mu 1_{qr}^t + 1_s^t \otimes \beta + 1_s \gamma^t] + E$	(iv) $Y = X[\mu 1_{qr}^t + \alpha \otimes 1_r^t + 1_s \gamma^t] + E$	(v) $Y = X[\mu 1_{qr}^t + 1_q^t \otimes \beta + \alpha \otimes 1_r^t] + E$
Hypothesis assumed true	H_0^{3w}, H_0^{sq} and H_0^{sr}		H_0^{3w}, H_0^{sq} and H_0^{qr}
Model	(vi) $Y = X[\mu 1_{rq}^t + 1_q \gamma^t] + E$		(vii) $Y = X[\mu 1_{rq}^t + 1_s(1_q^t \otimes \tau_r^t) + \alpha \otimes 1_r^t] + E$
Hypothesis assumed true	H_0^{3w}, H_0^{sq} and H_0^{qr}		
Model	(viii) $Y = X[\mu 1_{rq}^t + 1_q^t \otimes \beta + 1_s(\tau_q^t \otimes 1_r^t)] + E$		
Hypothesis assumed true	Test for equal sector means given H_0^{3w}, H_0^{sq} and H_0^{sr} are true		Test for equal assessor means given H_0^{3w}, H_0^{sq} and H_0^{qr} are true
Hypothesis assumed true	$H_0^t : C_s \theta 1_{rq} = 0_{(s-1) \times 1}$		$H_0^t : 1_s^t \theta [1_q \otimes C_r] = 0_{1 \times (r-1)}$
Model	(ix) $Y = X[\mu_c 1_s 1_{qr}^t + 1_s \gamma^t] + E$		(x) $Y = X[\mu 1_{rq}^t + \alpha \otimes 1_r^t] + E$
Hypothesis assumed true	Test for equal question(task) means given H_0^{3w}, H_0^{sq} and H_0^{qr} are true		
Model	$H_0^t : 1_s^t \theta [C_q \otimes 1_r] = 0_{1 \times (q-1)}$ (xi) $Y = X[\mu 1_{rq}^t + 1_q \otimes \beta] + E$		

1_n is a $n \times 1$ vector of ones; $0_{n \times m}$ is a $n \times m$ matrix of zeros, and $C_n = \begin{pmatrix} I_{n-1} \\ 1_{n-1} \end{pmatrix}$.

Table 1 (continued)

Assume no interactions exist between factor and response variables		
Model	(xii)	$Y = X[\mu 1_{qr}^t + 1_s(\tau_q^t \otimes 1_r^t) + 1_s(1_q^t \otimes \tau_r^t)] + E$
Hypothesis	$H_0^t : C_s \theta 1_{qr}^t = 0_{s-1}$	$H_0^t : 1_s^t \theta [C_q \otimes 1_r^t] = 0_{q-1}^t$ $H_0^t : 1_s^t \theta [1_q^t \otimes C_r] = 0_{r-1}^t$
Model	(xiii) $Y = X[\mu 0 1_{rq} + 1_s(1_q^t \otimes \tau_r^t) + 1_s^t(\tau_q^t \otimes 1_r^t)] + E$	(xiv) $Y = X[\mu 1_{qr}^t + 1_s \otimes (1_q^t \otimes \tau_r^t)] + E$ (xv) $Y = X[\mu 1_{qr}^t + 1_s(\tau_q^t \otimes 1_r^t)] + E$
Hypotheses assumed true	H_0^t and H_0^t are true	
Model	(xvi) $Y = X[\mu 0 1_{rq} + 1_s(\tau_q^t \otimes 1_r^t)] + E$	(xvii) $Y = X[\mu 1_{rq}^t] + E$
Hypotheses assumed true	H_0^t and H_0^t	
Model	(xviii)	$Y = X[\mu 0 1_{rq} + 1_s(\tau_q^t \otimes 1_r^t)] + E$
Hypotheses assumed true	H_0^t ; H_0^t and H_0^t	
Model	(xix)	$Y = X[\mu 0 1_{rq} + 1_s(\tau_q^t \otimes 1_r^t)] + E$

To save space in the above table and the paper, the (approximating) models records lower order effects only if higher order interactions are not present.

Definition of the location parameters of the models:

μ_0 pooled sector mean effect;

μ is an $s \times 1$ vector of sector means;

τ_r is a $r \times 1$ vector of assessor (rater) means;

τ_q is a $q \times 1$ vector of question means;

γ is a $rq \times 1$ vector of parameters that indicate the influence the interaction between assessors and questions have in the model;

β is a $s \times r$ matrix of parameters that indicate the influence of sector-assessor interaction;

α is a $s \times q$ matrix of parameters that indicate the influence of sector-question interaction;

θ is the $s \times rq$ matrix that contains the influences sector-question-assessor 3-way interaction , 2-way interaction and individual mean effects.

$C'_q \theta (1_q \otimes C_r) = 0$. Accordingly if H_0^{3w} is assumed true, then H_0^{sq} , H_0^{sr} and H_0^{qr} can be expressed as $C'_s \alpha C_q = 0$, $C'_q \beta C_r = 0$ and $\gamma'(C_q \otimes C_r) = 0$, respectively.

The classical multivariate test statistics are summarised in Table 2. References for approximate percentage points are made; exact percentage values are unknown for the design considered.

Table 2

Classical test statistics and their approximate distributions for hypotheses of the form $H_0 : C\theta(V_1 \otimes V_2) = 0$

$$H = [C\hat{\theta}(V_1 \otimes V_2)]'[C(X'X)^{-1}C'] [C\hat{\theta}(V_1 \otimes V_2)]$$

$$E = n^2 V_1' Q_{mle} V_1 \otimes V_2' R_{mle} V_2.$$

Let the rank of matrices C , V_1 and V_2 be k , m_1 and m_2

Test Statistics	Reference	Approximate distribution
$(n-s)tr(E^{-1}H)$	Generalised Hotellings T^2	$\chi^2_{km_1m_2}$
$n \cdot \ln I + E^{-1}H $	Wilk's Criterion	$\chi^2_{km_1m_2}$
$n \cdot tr[(E + H)^{-1}H]$	Pillai's Criterion	$\chi^2_{km_1m_2}$

3. THE APPROACH TO MODEL SELECTION

Model selection has been applied to analysis of variance problems [see Linhart and Zucchini (1986)]. An approach similar to Linhart and Zucchini's will now be extended to the MANOVA and GMANOVA. The multivariate criteria used to select models will be identical to the $C_p^{ur}(\Sigma)$ criterion discussed in Sparks (1987). Although Sparks (1987) uses this criterion to select variables in the seemingly unrelated regression model, it may also be used to select approximating models to operating models that can be written in the regression form of equation (1). Let p be the number of estimable location parameters in the approximating model, then the $C_p^{ur}(\Sigma)$ criterion can be expressed as

$$C_p^{ur}(\Sigma) = qr(n-s)tr[E^{-1}H] + 2p - sqr$$

Models with $C_p^{ur}(\Sigma)$ close to p are good models [for more details on the interpretation of this criterion see Sparks (1987)]. It is convenient to transform this criterion to a measure which is invariant of p for easy interpretation. Such a transformation is available in Sparks et al (1983) page 1786, equations (5.4), namely

$$T_p^* = 1 + (C_p^{ur}(\Sigma) - p)/(sqr - p).$$

Models which have T_p^* values close to 1 are classified as good approximations to the operating model. A backward elimination (BE) approach is used to decide on the "best" approximating model (empirically defined as the model with the smallest T_p^* value). The path the backward elimination will take corresponds closely to the nested hypotheses outlined in Table 1. The BE steps are outlined below:

Step 1: Decide whether sectors, tasks and assessors jointly interact i.e. check whether model (i) can be approximated by (ii) in Table 1. If there is a three way interaction then stop and conclude that no approximating model recorded in Table 1 is "better" than the operating model and hence there is a three way interaction between sectors, tasks and assessors. In such situations it is extremely difficult to achieve realistic and unbiased assessments of performances in the population when performing these tasks. If approximating model (ii) is "better" than the operating model (i), (i.e. if $T_p^* \leq 1$) then proceed to Step 2;

Step 2: Decide whether there are any two-way interactions e.g., check whether model (i) can be "better" approximated by models (iii), (iv), (v), (vii) or (viii). For example, if only sectors and assessors interact then model (viii) is a "better" model than (ii). If more than one two-way interactions provide information about the student scores then stop at this stage and choose the appropriate model from (iii), (iv) or (v), i.e. select the model with the smallest T_p^* value. Otherwise proceed to Step 3;

Step 3: This step is split into two parts depending on whether any variables are found to interact in Step 2. If no variables interact then

proceed to part (a) otherwise go to (b).

(a) Decide whether either sectors, tasks and assessors have different mean performance scores. For example, decide whether models (xiii), (xiv) or (xv) are "better" than (xii). In turn decide whether these can be reduced to either models (xvi), (xvii) or (xviii) without loss of information. The simplest approximating model that can be fitted is model (xix);

(b) Decide whether the mean parameters of the variable/factor that failed to interact with the other variables/factor(s), need to be fitted. For example, if sectors fail to interact with the other variables then decide whether model (xi) is a "better" approximation to the operating model than (vi).

4. A TEST PROCEDURE FOR MODERATE SAMPLE SIZES

Let $\psi(E^{-1}H)$ denote a function of the eigenvalues of matrix $E^{-1}H$. The exact distribution of $\psi(E^{-1}H)$ is unknown although asymptotic tests can be used for large data sets. An alternative procedure applicable to smaller data sets will now be proposed. The normal assumption is made in the procedure to follow, however this can easily be extended to other distributions.

Let the bootstrap estimate of the null distribution of Y be

$$N(X\hat{\theta}_{RH}; I_n \otimes Q_{mle} \otimes R_{mle})$$

where: $\hat{\theta}_{RH}$ is a (restricted) least squares estimate of θ subject to the constraint $C\hat{\theta}(V_1 \otimes V_2) = 0$.

Restrictions may sometimes be necessary to make the regression coefficients uniquely estimable. This bootstrap distribution is estimated using all original data values. The bootstrap population (BP) is used to generate samples of Y , denoted by Y_{BS} . These bootstrap samples (BS) are used to estimate the distribution for the test statistics under the null hypotheses (Sparks, 1988), which in turn can be used to test the hypothesis. For example, let $\psi(E^{-1}H)$ (a function of the eigenvalues of $E^{-1}H$) be the observed value for the test statistic using the BP. The classical p -values

can be taken as

$$p = Pr[\psi(E_{BS}^{-1}H_{BS}) \geq \psi(E^{-1}H)].$$

The method of finding these p -values is outlined in Sparks, 1988, and hence is not covered here.

5. A DESCRIPTIVE ANALYSIS OF THE DATA

A descriptive technique for analysing this application can be useful for researchers who find procedures MANOVA and GMANOVA difficult to interpret. The technique proposed in this paper is the covariance biplot (Gabriel, 1972). This procedure anticipates a good low dimensional fit to the data, as it is expected that the assessor ratings are highly correlated and it is also anticipated that scores for related questions are correlated within students. Accordingly the covariance biplot should produce a reasonable fit of the data. This procedure is briefly outlined below.

If the data matrix

$$Y_c = Y - 1_n 1'_n Y/n$$

can be represented in two dimensions without losing too much information then the covariance biplot will prove useful. Let the singular value decomposition of Y_c be

$$Y_c = U D_\lambda V'$$

such that $V'V = I_{qr}$, $U'U = I_n$ and $D_\lambda = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_{qr} \end{pmatrix}$, where

$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{qr}$. A good two dimensional representation of Y_c will be found if $\sum_{j=3}^{qr} \lambda_j / (\sum_{j=1}^{qr} \lambda_j) \approx 0$. As such

$$(7) \quad Y_c \approx U_1 D_{\lambda_1} V_1'$$

where: U_1 and V_1 denote $n \times 2$ and $qr \times 2$ matrices consisting of the first two columns of U and V respectively.

The biplot is a two dimensional plot of the rows $H = U_1$ and $G = V_1 D_{\lambda_1}$ (as coordinates). Note that from (6):

$$Y'_c Y_c \approx GG'.$$

Let g'_i and h'_i define the i 'th row of G and H respectively, then the following interpretations can be assigned to the biplot:

- (i) $g'_i g_j$ is an approximate estimate of $(n-1)\sigma_{ij} + \theta'_i X'_c X_c \theta_j$ where θ'_i is the i 'th row of θ ;
- (ii) $h'_i g_j$ is approximately equal to the i 'th row and j 'th column of Y_c .

If points h_i and g_j are in the same quadrant and both far from the origin then the i th student performance score in column j is well above the average for column j . If points h_i and g_j are in opposite quadrants then the i th student performance score in column j is below the average.

If the covariances are of prime interest, then the effects of the location parameters can be removed by analysing the biplot of matrix $Y_c = Y - X\hat{\theta}$ where $\hat{\theta}$ is an estimate of θ . As such $g'_i g_j$ will approximately estimate $(n-s)\sigma_{ij}$, and $g'_i g_j / (\|g_i\| \|g_j\|)$ [i.e. the cosine of the angle between vectors g_i and g_j] will approximately estimate ρ_{ij} (the correlation between the i 'th and j 'th response variable). These can be used to interpret the scale of the student scores after the effects of the location parameters have been removed. If the "true" model of the assessment scores included no interaction terms, then for assessor i and assessor j note that $X_c \theta_m = X_c \theta_\ell$ and hence $\theta'_m X'_c X_c \theta_m = \theta'_\ell X'_c X_c \theta_\ell$ where $m = (k-1) \times r + i$, $\ell = (k-1) \times r + j$ for all $k = 1, \dots, q$, $i, j = 1, \dots, r$ and $i \neq j$. As such the relative positioning of the within question g_i values for the biplot of $Y_c = Y - 1_n 1'_n Y/n$ and the corresponding g_i values of biplot $Y_c = Y - X\hat{\theta}$ should be approximately similarly orientated and proportionately distanced from the origin in both biplots. In similar studies to that presented in this paper one could use a comparison of these biplots to explore interactions between variable(s) and/or factor(s).

6. EXAMPLES OF PLOTTED PROFILES AND THEIR INTERPRETATION

In order to illustrate the aims of this study, examples of interactions and the plots of their profiles are presented in Table 3. This presents figures which plot θ_{jkl} , the operating models location parameter for the j 'th sector of the population, k 'th question and ℓ 'th assessor, on the vertical axis in order of magnitude against the assessor and question number on the

horizontal axis. For convenience, the vertical axis is not displayed in these figures. The examples assume that there are two sectors of the population ($j = 1, 2$), two questions ($k = 1, 2$) and four assessors ($\ell = 1, 2, 3, 4$). The three way interaction example is not easy to interpret, however, examples of two way interactions presented in Table 3 will now be discussed. The example illustrating question-sector interaction, is interpreted as students from sector 1 prefer question 1 relative to students from sector 2. This indicates a cultural bias in questions, e.g. question 2 may require background knowledge that is foreign to the cultural background of students from this sector. The example of sector-assessor interaction indicates that assessor 3 assigns on average lower scores to students from sector 1 relative to other assessors, thus indicating that this assessor discriminates against students from sector 1. The example of question-assessor interaction indicates that assessor 3 assigns higher scores to question 2 than expected when compared with other assessors. The example of assessor's means being equal indicates that students from sector 1 perform better on average in question 2 than in question 1. The example of equal question mean scores indicates that assessor 1 discriminates against students from sector 1 and in favour of students from sector 2. The example of equal sector means indicates that the students from the two sectors of the population are equally able in terms of their performance scores to these two questions, however, assessor 2 assigns relatively lower scores on average to student performances in question 2 than those in question 1.

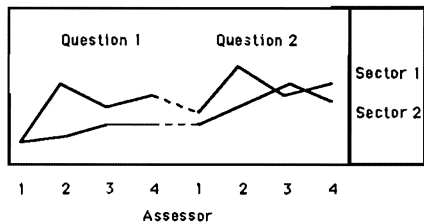
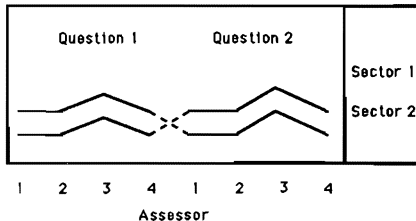
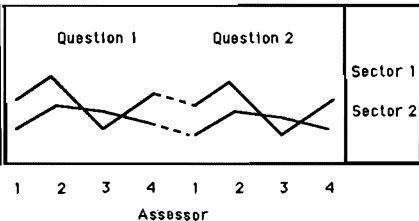
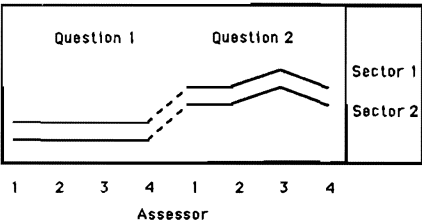
7. AN EXAMPLE OF APPLICATION

In order to illustrate the above methodology, examination papers for first year students (target population) in the Department of Environmental and Geographical Science at the University of Cape Town were rated by six assessors. The aim of the study was to determine whether:

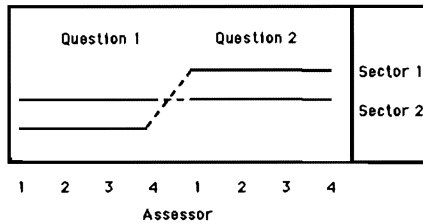
- (i) certain cultural groups score more highly on some questions than others;
- (ii) assessors exhibit relative bias in their assessment of questions drawn from two different cultural groups;

Table 3: Examples of interactions and the plots of their profiles

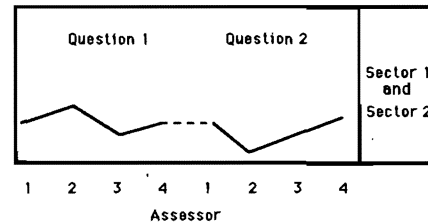
The plots below have θ_{jkl} on the vertical axis for questions $k = 1, 2$; sectors $j = 1, 2$ and assessors $l = 1, 2, 3, 4$. The axes however have been omitted from the plots.

An Example of a three way interaction**Examples of two way interactions when it is known that only one pair of factors and/or variables interact****An interaction between question and sectors****An interaction between sectors and assessors****An interaction between questions and assessors**

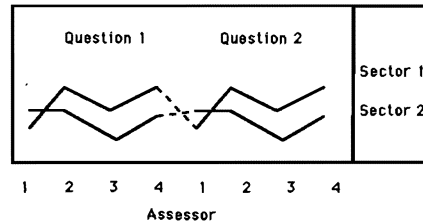
Questions and sectors interact but assessors have equal means



Questions & assessors interact but sectors have equal means



Sectors and assessors interact but questions have equal means



- (iii) interactions exist between assessors and particular questions;
- (iv) the cultural groups are on average perceived as performing at a similar academic level for all questions;
- (v) question mean scores indicate relative differences in question difficulty;
- (vi) assessors on average employ similar assessment strategies.

The examination structure in the Department of Environmental and Geographical Science at the University of Cape Town requires students to write three different essay questions. Examination essays were randomly selected for twenty students in the target population of which twelve were from "black" and eight "white", candidates. Names were removed from the essay scripts in order to ensure that assessors would not immediately be able to identify the race of the candidates. This was necessary as it is possible in multi-cultural environments for assessors "unconsciously" to compensate for, or discriminate against, the ideas and work of certain groups. In South Africa, the potential for such increases as assessors have grown up in largely racially segregated institutions and may have varying academic expectations, standards and assessment strategies.

The six assessors were post-graduate students within the Department of Environmental and Geographical Science at the University of Cape Town. As far as possible assessors were chosen for their differing educational and cultural backgrounds.

Let y_{ijkl} be as defined in section 2. Tasks are the three essay questions and sectors of the population are "white" and "black" students. The operating model is given by (1) with $q = 3$; $r = 6$; $n_1 = 12$; $n_2 = 8$ and $n = 20$.

The nested hypotheses outlined in Table 1 are now tested using the bootstrap approach outlined in Section 4 and the p -values for the appropriate tests are recorded in Table 4. These produced only one significant interaction, namely that between questions and assessors (p -value = 0.068; 0.066 or 0.002 depending whether the determinant, trace or the largest root is used as ψ). The plot of the estimated profiles in Figure 1 indicates this

Figure 1: Plot of the profiles for questions after averaging over sectors

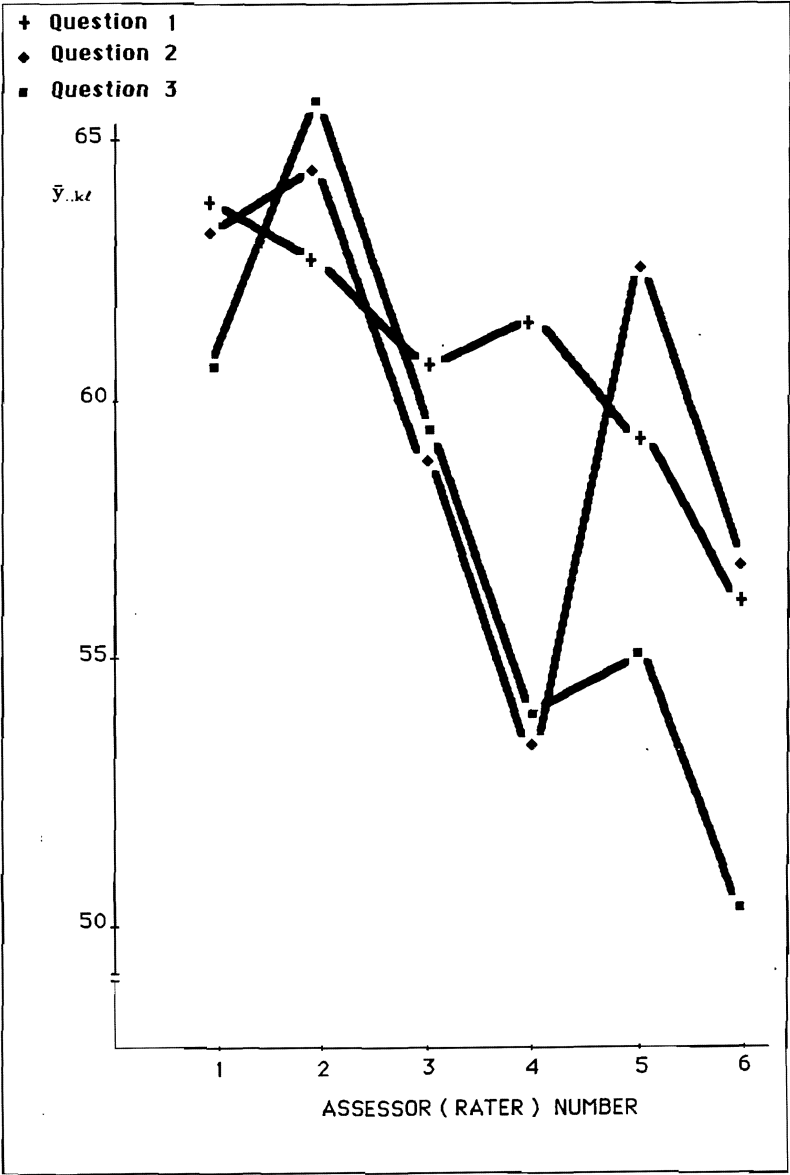


Figure 2: The Covariance Biplot of $Y_c = (I_n - 1_n 1_n/n)Y$

The students are represented by letters a to t. 'Blacks' are labelled a to l, while 'Whites' are m to t.

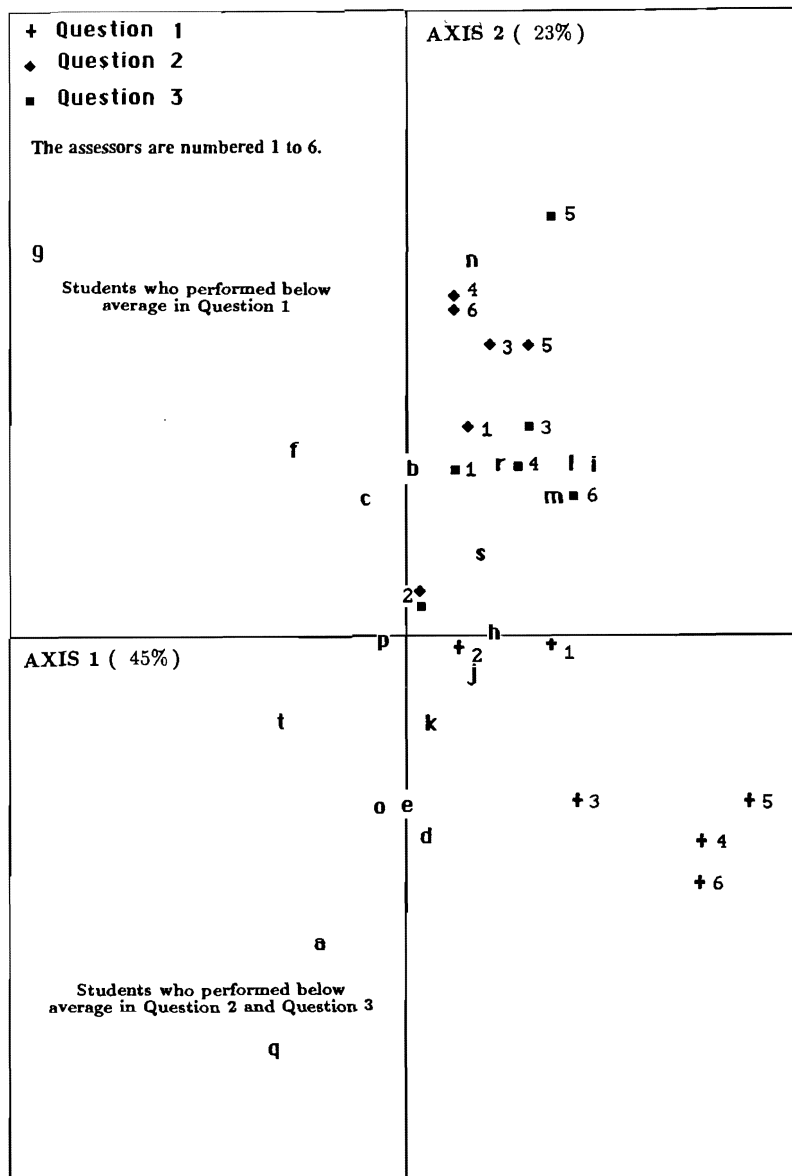
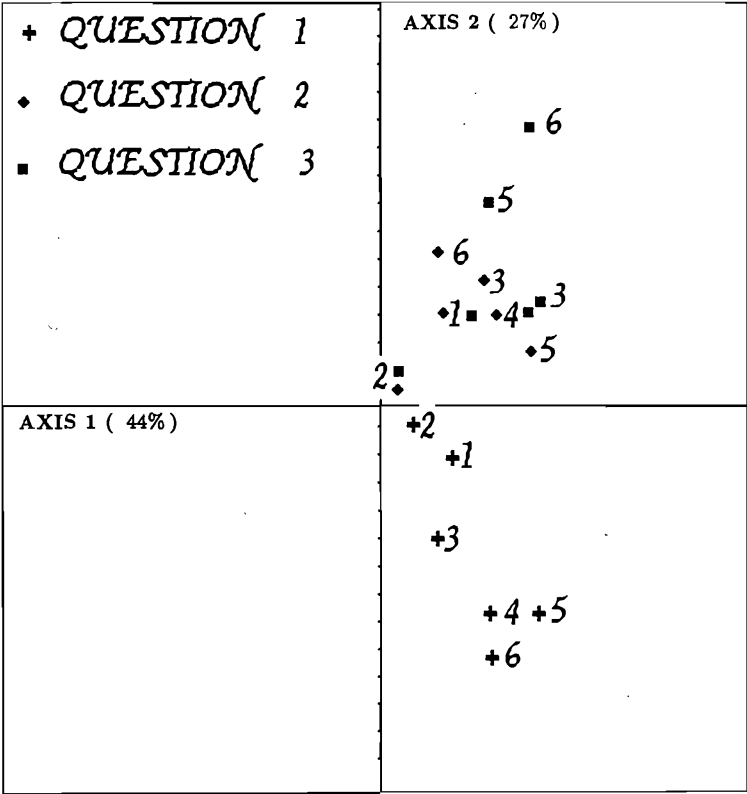


Figure 3: The Covariance Biplot of $Y_c = Y - X\hat{\theta}$

The numbers indicate the raters e.g. ♦5 indicates the position of the 5th rater when assessments were made to student answers to question 2.



interaction. This is interpreted as certain assessors assigning on average higher(lower) marks relative to others for particular questions. For example, assessor 4 assigns much higher scores to question 1, and Assessor 6 much lower marks to question 3, relative to other assessors. Assessor 2 is also found to assign lower scores on average to question 1 relative to other question-assessor combinations.

Table 4
p-values and criterion values of model section

A	$p\text{-value} = \hat{P}(\psi(E_{BS}^{-1}H_{BS}) \geq \psi(E^{-1}H/H_0^A))$					T_p^*
	ψ	Determinant	Trace	Largest Root	p	
3w			0.643		26	0.805
sq			0.499		24	0.933
sr			0.334		21	1.001
qr		0.068	0.066	0.002	16	1.690
s			0.989		18	0.000

(E_{BS} and H_{BS} are the error and hypothesis matrices established from the Bootstrap sample)

The fact that no significant interaction was found between assessors and sectors ($p\text{-value} = 0.334$), indicates that these assessors do not exhibit a significant bias for or against any sector of the population. No significant interaction between sectors and questions ($p\text{-value} = 0.499$) indicates that the questions are not biased in favour of students from any one sector e.g. "black" students do not perform on average better than "whites" on question 1 relative to question 2.

The test of hypothesis H_0^s is feasible since sectors do not interact with any of the response variables. This test shows no significant difference in the ability of "white" and "black" students in terms of their average performance scores ($p\text{-value} = 0.989$).

The criteria values for the appropriate approximating models are recorded in Table 4. These suggest that several approximating models are "better" than the operating model. The model with the smallest number of location

parameters approximating the operating model in terms of the selection criteria is

$$y_{ijkl} = \mu_0 + \gamma_{kl} + e_{ijkl} \text{ i.e. } \theta = \mu_0 \underline{1}_s \underline{1}'_{gr} + \underline{1}_s \underline{\gamma}'.$$

This is assumed to be the "best" approximating model since all redundant explanatory information has been excluded. The estimated coefficients are recorded in Table 5. This confirms some of the observations in Figure 1, namely assessor 2, 4 and 6's behaviour. Note that Assessor 1 and 2 always assign more marks on average than other assessors (i.e. $\gamma_{ij} > 0$ for all $j = 1, 2, 3$ and $i = 1$ and 2).

Table 5

An estimate of the regression coefficients for the approximating model

Question 1						
Coefficient	γ_{11}	γ_{12}	γ_{13}	γ_{14}	γ_{15}	γ_{16}
Estimate	3.85	2.45	1.00	1.95	0.15	-2.65
Question 2						
Coefficient	γ_{21}	γ_{22}	γ_{23}	γ_{24}	γ_{25}	γ_{26}
Estimate	3.60	5.75	-1.20	-5.15	2.60	-2.45
Question 3						
Coefficient	γ_{31}	γ_{32}	γ_{33}	γ_{34}	γ_{35}	γ_{36}
Estimate	1.40	8.40	-0.50	-5.05	-4.15	-10.00

A descriptive analysis was carried out by examining the covariance biplot of

$$Y_c = (I_n - \underline{1}_n \underline{1}'_n / n) Y \text{ and } Y_c = Y - X \hat{\theta}$$

(see Figure 2 and Figure 3, respectively). The biplot explained, respectively, 68% and 71% of the total variation. Figure 3 indicates clearly that question 1 assessed scores are approximately independent of question 2's and question 3's. This is confirmed when Q_{mle} is examined. The estimated correlations between questions 1 & 2; 2 & 3 and 1 & 3 were 0.032; -0.055 and -0.004 respectively. The within question assessments of each assessor were highly related, however, it is quite clear that assessor's scores

differ in spread (variance). It is also noticeable that the spread of scores for question 2's assessments is lower on average when compared to the other two questions. This could be interpreted in a number of ways, one being that question 2 is not as good a discriminator of student performances as the other two questions. Figure 2 gives extra information which mainly consists of the relationship of students' performances with the question-assessor combinations.

8. CONCLUDING REMARKS

Analyses involving interactions between questions, assessors and/or sectors of a population is necessary in a multicultural environment. In order to make assessment fair it is necessary to know about the interactions involved in the assessment of student performances. For example, setting probability questions relating to card games may produce lower or zero marks for students from a particular sector of the population, as such games are unknown to that sector.

The bootstrap approach can be applied without any distributional assumptions. Note that the predicted values of the response variables at each observed point is given by

$$\begin{aligned}\hat{Y} &= X\hat{\theta} + (\hat{Y} - X\hat{\theta}) \\ &= X\hat{\theta} + \hat{E}\end{aligned}$$

Let $\hat{E}$ be partitioned as follows

$$\hat{E}' = (\hat{E}'_1 \dots \hat{E}'_s)$$

where $\hat{E}_i$ are the ordinary residuals corresponding to the i 'th sector of the population, i.e. $\hat{E}_i$ is $n_i \times qr$. Let E_i^* be a $n_i \times qr$ matrix made up by selecting randomly with replacement n_i rows from $\hat{E}_i$. A bootstrap sample of the response variables is hence given by

$$Y^* = X\hat{\theta} + \hat{E}_i$$

Repeated bootstrap samples of Y can be used to estimate the distribution of the test statistics and hence find p -values.

The bootstrap procedure allows greater scope for applying more appropriate test statistics than just the determinant, trace and largest root as options for function ψ . For example, if 99% of the "information" in $E^{-1}H$ is contained within the first two eigenvalues of this matrix, then find the bootstrap distribution of any appropriate function of these two eigenvalues and use this to test the hypothesis of interest. The bootstrap procedure also has the advantage of being able to estimate the distribution of $\psi(E^{-1}H)$ when it becomes necessary to constrain the covariance structure before testing hypotheses. This is particularly useful when the tests relating to the location parameters are solely of interest and there are insufficient observations to estimate the full covariance structure. For example, for small data sets, a test procedure which constrains all correlations to be equal before estimating the test statistic, may be powerful at detecting deviations from the null hypothesis. The bootstrap procedure is particularly useful when it is known that the data (or any simple transformation of it) are not normal. These areas have sound practical implications and need attention.

REFERENCES

- GABRIEL, K.R. (1972). Analysis of Meteorological Data by Means of Canonical Decomposition and Biplots. *J. App. Meteor.* **11**, 1071-1077.
- LINHART, H. and ZUCCHINI, W. (1986). *Model Selection*. John Wiley and Sons, New York.
- MARDIA, K.V., KENT, J.T. and BIBBY, J.M. (1979). *Multivariate Analysis*. Academic Press, New York.
- MORRISON, D.F. (1990). *Multivariate Statistical Methods* (3rd ed.), McGraw-Hill, New York.
- SPARKS, R.S., COUTSOURIDES, D. and TROSKIE, L. (1983). The Multivariate Cp. *Commun. Statist. Theor. Meth.*, **A12**, 1775-1793.
- SPARKS, R.S. (1987). Selecting Estimators and Variables in the Seemingly Unrelated Regression Model. *Commun. Statist. Comp. Sim.*, **B16**, 99-127.
- SPARKS, R.S. (1988). A distribution free method of detecting outliers in regression. *Commun. Statist. Theor. Meth.*, **A17**, 705-727.

SRIVASTAVA, M.S. and CARTER, E.M. (1983). *An Introduction to Applied Multivariate Statistics*. Elsevier North-Holland, New York.

SRIVASTAVA, M.S. (1987). Profile Analysis of Several Groups. *Commun. Statist. Theor. Math.*, **A16**, 909-926.

APPENDIX A

Let σ_{ij} denote the i th row and j th column element of matrix Σ and $tr(A)$ the trace of square matrix A . Consider the following properties of multivariate calculus:

P1: Suppose that Σ is a symmetric matrix, then

$$\frac{\partial \ln|\Sigma|}{\partial \sigma_{ij}} = (2 - \delta_{ij})[\Sigma^{-1}]_{ij}$$

where:

$$\begin{aligned}\delta_{ij} &= 1 \text{ if } i = j \\ &= 0 \text{ if } i \neq j.\end{aligned}$$

Proof

Denote J_{ij} by the matrix with all elements equal to zero with the exception of the i th row and the j column which is equal to one. Let $[\Sigma^{-1}]_{ij}$ denote the i th row and j th column element of Σ^{-1} ; then

$$\begin{aligned}\frac{\partial \ln|\Sigma|}{\partial \sigma_{ij}} &= tr \left[\frac{\partial \ln|\Sigma|}{\partial \Sigma} \frac{\partial \Sigma}{\partial \sigma_{ij}} \right] \text{ chain rule} \\ &= tr[\Sigma^{-1}(J_{ij} + (2 - \delta_{ij})J_{ji})] \\ &= (2 - \delta_{ij})[\Sigma^{-1}]_{ij}.\end{aligned}$$

P2: Suppose that Σ is a symmetric matrix. Then

$$\frac{\partial \Sigma^{-1}}{\partial \sigma_{ij}} = -\Sigma^{-1}(J_{ij} + (2 - \delta_{ij})J_{ji})\Sigma^{-1}.$$

Proof

Consider the partial derivative of both sides of the equation $\Sigma^{-1}\Sigma = I$ with respect to σ_{ij} . This leads to

$$\frac{\partial \Sigma^{-1}}{\partial \sigma_{ij}}\Sigma + \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_{ij}} = 0.$$

Post-multiplying both sides of the above equation by Σ^{-1} leads to

$$\frac{\partial \Sigma^{-1}}{\partial \sigma_{ij}} + \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_{ij}} \Sigma^{-1} = 0,$$

from which **P2** follows.

Derivation of the Maximum Likelihood Estimator of R and Q

The likelihood function for the proposed model is

$$L(R, Q, \beta | Y, X) = \frac{1}{(2\pi)^{\frac{1}{2}nqr} |R|^{\frac{1}{2}nq} |Q|^{\frac{1}{2}nr}} + \exp\left(-\frac{1}{2} \text{tr}[Q^{-1} \otimes R^{-1}]G\right).$$

Hence, the log-likelihood is

(a)

$$\begin{aligned} \ln L(R, Q, \beta | Y, X) &= -\frac{1}{2}nqr \ln(2\pi) - \frac{1}{2}nq \ln(|R|) \\ &\quad - \frac{1}{2}nr \ln(|Q|) - \frac{1}{2} \text{tr}([Q^{-1} \otimes R^{-1}]G) \\ &= -\frac{1}{2}nqr \ln(2\pi) - \frac{1}{2}nq \ln(|R|) \\ &\quad - \frac{1}{2}nr \ln(|Q|) - \frac{1}{2} \sum_{k=1}^q \sum_{t=1}^q \text{tr}([Q^{-1}]_{kt} R^{-1} G_{tk}). \end{aligned}$$

The derivative of this log-likelihood with respect to r_{ij} , the i th row and j th column element of matrix R , is

$$\begin{aligned} \frac{\partial \ln L(R, Q, \beta | Y, X)}{\partial r_{ij}} &= -\frac{1}{2}nq(2 - \sigma_{ij})[R^{-1}]_{ij} + \\ &\quad \frac{1}{2} \sum_{k=1}^q \sum_{t=1}^q \text{tr}([Q^{-1}]_{kt} R^{-1} (J_{ij} + (2 - \delta_{ij})J_{ji}) R^{-1} G_{tk}) \\ &= (2 - \delta_{ij})\left(-\frac{nq}{2}[R^{-1}]_{ij} + \frac{1}{2} \sum_{k=1}^q \sum_{t=1}^q ([Q^{-1}]_{kt} [R^{-1} G_{tk} R^{-1}]_{ij}\right) \end{aligned}$$

The equations formed by equating the above to zero for each $i = 1, \dots, r; j = 1, \dots, t$ can be expressed as follows:

$$-\frac{nq}{2}R^{-1} + \frac{1}{2} \sum_{k=1}^q \sum_{t=1}^q ([Q^{-1}]_{kt} R^{-1} G_{tk} R^{-1}) = 0.$$

Given matrix Q and matrix β , the maximum likelihood estimator of R is

(b)

$$R_{mle} = \frac{1}{nq} \sum_{k=1}^q \sum_{\ell=1}^q [Q^{-1}]_{k\ell} G_{\ell k}.$$

Note that equation (a) can be expressed as

$$\begin{aligned} \ln L(R, Q, \beta | Y, X) = & -\frac{1}{2} nqr \ln(2\pi) - \frac{1}{2} nq \ln(|R|) - \frac{1}{2} nr \ln(|Q|) \\ & - \frac{1}{2} \sum_{k=1}^q \sum_{\ell=1}^q [Q^{-1}]_{k\ell} \text{tr}(R^{-1} G_{\ell k}). \end{aligned}$$

If the ℓ th row and k th column element of matrix A is given by $\text{tr}(R^{-1} G_{\ell k})$ then the log-likelihood function can be expressed as

$$\ln L(R, Q, \beta | Y, X) = -\frac{1}{2} nqr \ln(2\pi) - \frac{1}{2} nq \ln(|R|) - \frac{1}{2} nr \ln(|Q|) - \frac{1}{2} \text{tr}[Q^{-1} A].$$

As such, it is easy to see that, given matrix R and matrix β , the maximum likelihood estimator of Q is

(c)

$$Q_{mle} = \frac{1}{nr} A.$$

The maximum likelihood estimator of θ is

(d)

$$\hat{\theta} = (X'X)^{-1} X'Y.$$

Using (b), (c) and (d), the maximum likelihood estimator of Q and R can be found iteratively.